Lab 1 – AI^2 Product Description

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Table of Contents

1. Introduction 3
2. Product Description 6
   2.1 Key Product Features and Capabilities 6
   2.2 Major Components (Hardware/Software) 8
3. Identification of Case Study 10
4. Glossary 12
5. References 14

List of Figures

Figure 1 Current Process Flow of manual invoice processing at ADS 5
Figure 2 AI^2 Major Functional Components Diagram 8

List of Tables

1. Introduction

Manual invoice processing remains prevalent across organizations, with approximately 90% of companies still relying on manual methods to handle their invoices (Jex, 2024). This process often involves staff manually opening emails, entering data, and verifying spreadsheets line by line. Such inefficiencies come at a significant cost—manual processing can range from $12 to $40 per invoice, compared to just $3 to $6 when automation is implemented (Parseur, 2025). For organizations managing large invoice volumes, these costs escalate rapidly.

Beyond financial inefficiency, manual processing also results in slower turnaround times and increased risk of human error. On average, manual invoice handling takes 10 to 15 days, while automation can reduce this time to only a few days (Parseur, 2025). Furthermore, between 1–3% of manually processed invoices contain errors, which can lead to strained vendor relationships, payment delays, and financial discrepancies (Parseur, 2025).

The problem also extends to the people performing these tasks. A recent survey found that 96% of Accounts Payable (AP) professionals report dissatisfaction with their existing tools (Tamaro, 2025), and a Gartner study revealed that 60% of accountants make multiple financial errors each month, often due to overextension and repetitive workloads (Gartner, 2024). When automation is adopted, productivity in invoice processing can improve by up to 80%, allowing employees to focus on higher-value work, reduce mistakes, and experience greater job satisfaction (Tamaro, 2025).

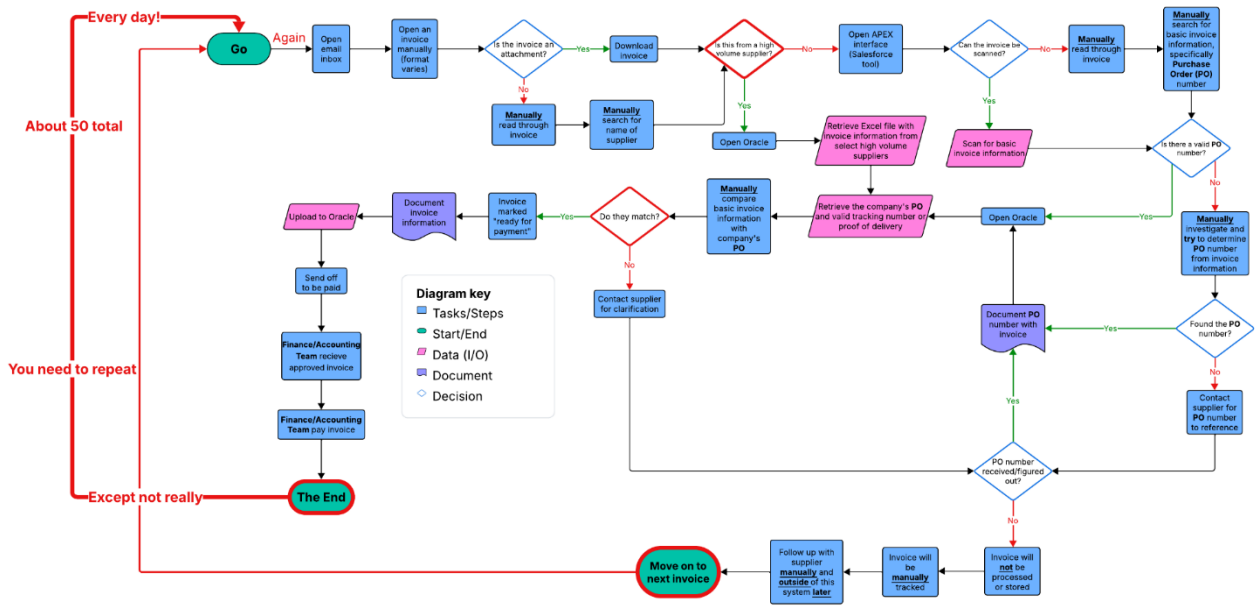
Manual invoice processing continues to place a substantial operational and financial burden on AP departments. As invoice volumes grow, these costs compound quickly, diverting significant staff time toward repetitive data entry, verification, and exception handling instead of higher-value financial analysis and strategic planning. The resulting workflow bottlenecks slow approvals,

reduce visibility into cash flow, and increase the likelihood of missed early-payment discounts and avoidable late fees. Over time, this inefficiency constrains scalability, limits an organization's ability to adapt to growth, and places unnecessary strain on accounting teams that could otherwise focus on improving financial performance and supporting broader business objectives.

Atlantic Diving Supply (ADS) is a large U.S.-based government and commercial supplier specializing in tactical equipment, operational gear, and logistics support for defense, federal, and public safety organizations. At ADS, these challenges are particularly evident. The company processes more than 150,000 invoices annually from over 3,000 suppliers, with invoices arriving in diverse formats such as PDFs, scans, and spreadsheets. A small AP team of six members must manually review each invoice and cross-check it against purchase orders (PO) in the system. This high workload leads to prolonged processing times, repetitive tasks, and employee fatigue, while also introducing the risk of errors and backlogs. Figure 1 illustrates the current manual workflow used by ADS's AP department.

Figure 1

Current Process Flow of manual invoice processing at ADS



The effects of this inefficiency extend beyond the AP team to ADS's leadership, vendors, and clients. Slow processing and payment delays strain supplier relationships and impact financial operations. To maintain strong partnerships and support sustainable growth, ADS—and organizations like it—require a solution that can automate the intake and processing of routine invoices while empowering staff to focus on exceptions and higher-value financial tasks.

To address the inefficiencies of manual invoice handling, Artificially Intelligent Invoices (AI²) is proposed as a secure, web-based platform that automates and streamlines the AP process. AI² integrates directly with an organization's existing financial systems, accelerating invoice processing while maintaining human oversight and regulatory compliance. The system leverages artificial intelligence to capture, interpret, and validate invoice data across diverse formats and vendors, reducing repetitive tasks and minimizing human error.

AI² automatically retrieves invoices from designated email inboxes, extracts key information, and structures it for efficient review. Each invoice is then compared to the corresponding PO within the organization's enterprise system, allowing routine matches to be completed instantly. When discrepancies occur, such as missing or mismatched data, the system flags the invoice for human review, learning from corrections to improve future accuracy. All actions are logged for auditability, ensuring transparency and compliance with financial and cybersecurity standards.

Beyond ADS, AI² is built to scale across organizations facing similar challenges. Its adaptable design supports new vendors, variable invoice formats, and integration with multiple enterprise systems, making it a flexible and sustainable solution for modern financial operations.

2. Product Description

AI² is a web-based invoice processing platform designed to automate and manage the AP workflow through an integrated, modular architecture. The system captures incoming invoices, extracts structured data using intelligent processing techniques, compares invoice information against corresponding PO, and presents results through a centralized review interface. By combining automated matching with human-in-the-loop validation, AI² ensures efficiency without sacrificing oversight. The platform is designed to integrate with existing enterprise systems, enabling scalable deployment across organizations with diverse financial infrastructures.

2.1 Key Product Features and Capabilities

AI² includes several core features designed to automate and streamline the AP workflow.

The first key feature is automated invoice retrieval. AI² collects incoming invoices through a centralized email inbox, eliminating manual downloading and file handling. In the case of ADS, invoices are retrieved securely from Outlook GCC using the Microsoft Graph API.

The invoice parser and data extraction engine converts unstructured invoice documents into structured data fields, including invoice number, vendor name, total amount, and invoice date. By automatically extracting this information, the system reduces manual data entry and improves processing efficiency.

Once structured data is available, the automated PO matching algorithm compares invoice information against existing enterprise POs. If a direct match is identified, the invoice is marked as ready for payment and forwarded to the Finance Team. If an exact match is not found, the system evaluates additional attributes such as vendor, date, and amount to identify potential matches. When no reliable match can be determined, the discrepancy detection and flagging module assigns the invoice to a designated AP team member for manual review. AI² also includes an automated reminder system to ensure unresolved invoices are addressed promptly, reducing processing delays.

A distinguishing capability of AI² is its confidence scoring and learning mechanism. During PO matching, the system calculates a confidence score based on the similarity of invoice and PO attributes. Human corrections are incorporated into model refinement, allowing the system to improve matching accuracy over time and adapt to new invoice formats or vendor variations.

Finally, the audit logging module records all invoice data, system decisions, and human interactions within a secure database. This ensures traceability, supports internal audits, and aligns with applicable compliance standards such as Federal Risk and Authorization Management Program (FedRAMP) and Department of Defense Risk Management Framework (DoD RMF).

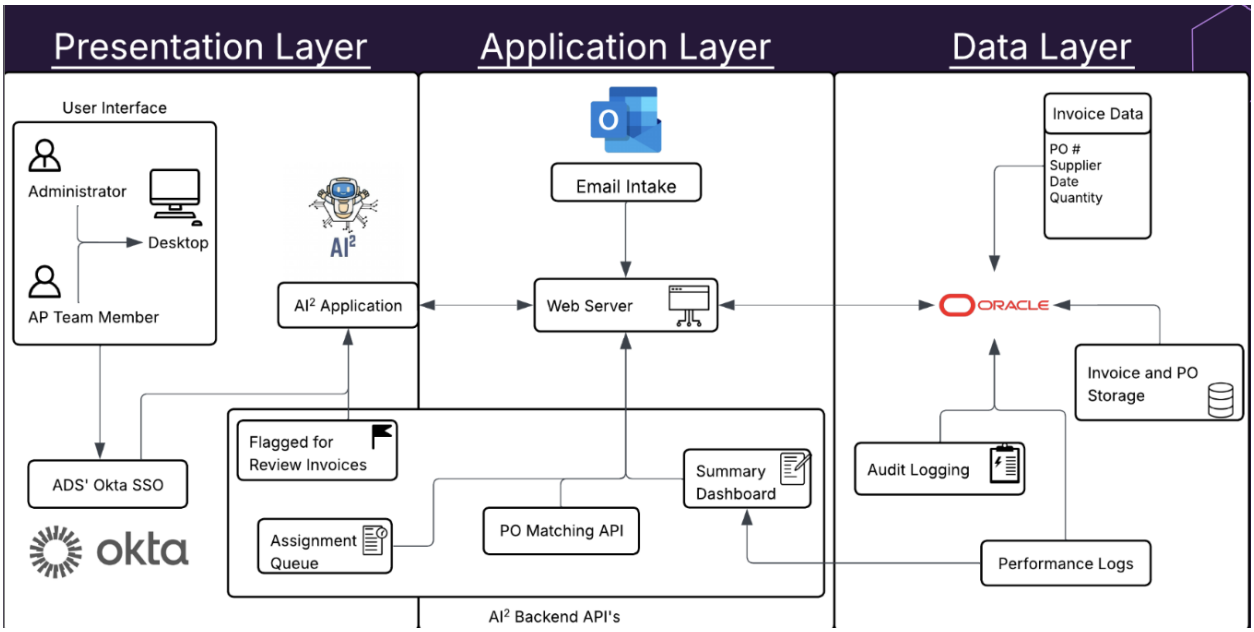
2.2 Major Components (Hardware/Software)

AI² is accessed through a web-based interface compatible with modern desktop browsers. End users, including AP team members and system administrators, interact with the platform via secure internet-connected workstations. The system is hosted within AWS GovCloud to satisfy federal security and compliance requirements, providing encrypted communication, secure storage, and controlled access. User authentication is handled through a Single Sign-On (SSO) solution. In the case of ADS, Okta SSO is used to ensure secure identity management and access control.

The major architectural components of AI² are organized into three primary layers, as shown below in Figure 2.

Figure 2

AI² Major Functional Components Diagram



The Presentation Layer consists of the AI² web application interface. This layer provides role-based access for administrators and AP team members. Administrators have access to system

health dashboards, performance monitoring tools, integration management panels, and queue rebalancing controls. AP team members access an assignment queue, flagged invoice review interface, and summary dashboard. This layer facilitates human oversight and interaction with automated system decisions.

The Application Layer contains the core business logic and processing modules. The Email Intake module retrieves invoices from designated email inboxes using secure API connections. The Web Server manages client requests and coordinates communication between the user interface and backend services. The PO Matching API executes invoice-to-purchase-order comparison logic and generates confidence scores. The Assignment Queue module routes flagged invoices to appropriate AP personnel, while the Summary Dashboard aggregates workflow statistics and system performance metrics.

The Data Layer manages persistent storage and logging. An Oracle database stores structured invoice data and PO records. Dedicated storage components maintain invoice and PO document records. The Audit Logging module records all system decisions, user interactions, and status changes to ensure traceability and regulatory compliance. Performance logs are maintained to monitor system health and operational efficiency.

The backend of AI² is developed using Flask, a lightweight Python web framework that manages routing, authentication, and business logic. Database operations are handled through SQLAlchemy, which provides an object-relational mapping (ORM) layer to interact with the underlying relational database. During development, SQLite is used as the database system due to its simplicity and ease of integration.

The frontend interface is styled using Tailwind CSS, enabling a responsive and modern user interface. The application follows a client-server architecture in which the Flask backend

processes requests, enforces logic, and communicates with the database, while the frontend is responsible for presenting data and handling user interactions.

3. Identification of Case Study

This case study examines how AI^2 is used within ADS by a small beta-testing group in the AP department. The group consists of two AP specialists responsible for invoice processing, one AP supervisor overseeing approvals and exceptions, and one IT liaison supporting system integration. These individuals are directly involved in the existing manual workflow and would serve as primary users of the prototype.

In a typical use scenario, incoming invoices are automatically ingested into AI^2, where key data fields are extracted and matched against existing purchase orders. Routine matches are processed automatically, while discrepancies are flagged for human review. Instead of manually reviewing every invoice, AP specialists focus only on exception cases, significantly reducing repetitive tasks and processing time. The supervisor accesses a dashboard displaying processing volume, match accuracy, and exception categories, improving visibility and oversight.

Through this beta test, the prototype demonstrates improved efficiency, reduced manual workload, increased accuracy in PO matching, and enhanced operational transparency. Feedback from this group would guide refinements prior to full-scale deployment.

Although this case study centers on ADS, high-volume manual invoice processing is common across government contractors, public agencies, healthcare systems, and large commercial distributors. A full-capability implementation of AI^2 would include expanded Enterprise Resource Planning (ERP) integration, advanced analytics, role-based access controls, and adaptive learning to improve performance over time.

Beyond ADS, the system could scale to support any organization managing large volumes of PO-based invoices, positioning AI² as a broadly applicable solution to systemic inefficiencies in financial operations.

4. Glossary

AI² (Artificially Intelligent Invoices): The proposed automated invoice processing system designed to streamline Accounts Payable workflows through AI-driven matching, classification, and routing.

Alteryx: A data automation and analytics tool that enables secure API connections and data workflows between applications.

AP (Accounts Payable): The amount a business owes for goods and services purchased on credit; typically, due at intervals of 30, 45, 60, or 90 days depending on repayment terms.

APEX: A strongly typed, object-oriented programming language that Salesforce developers use to execute flow and transaction control statements on the Salesforce platform.

Atlantic Diving Supply (ADS): An American federal contractor company that provides equipment and logistics solutions to the Department of Defense, federal agencies, and first responders.

AWS GovCloud: A secure U.S. government-compliant cloud environment that supports sensitive data and regulated workloads in accordance with FedRAMP and DoD requirements.

CMMC (Cybersecurity Maturity Model Certification): A framework developed by the U.S. Department of Defense to assess and enhance the cybersecurity posture of contractors handling federal information.

ERP (Enterprise Resource Planning System): An integrated software system used to manage core business processes, such as finance, procurement, and supply chain operations.

FedRAMP (Federal Risk and Authorization Management Program): A U.S. government program that standardizes security assessment, authorization, and continuous monitoring for cloud services.

Graph API: A Microsoft API interface used to access and automate data operations within Outlook and other Microsoft 365 services.

Invoice: A document listing goods or services provided, including a statement of the amount due; a bill.

Object-relational mapping (ORM): a programming technique that acts as a software layer translating data between relational databases and object-oriented programming languages.

Oracle: The system of record used by ADS for enterprise data and financial management.

PO (Purchase Order): The official order list ADS sends to suppliers detailing the items or services requested.

Salesforce: A robust Customer Relationship Management (CRM) platform that allows businesses to manage customer relationships efficiently.

Single Sign-On (SSO): an authentication method that allows users to securely access multiple, connected applications and websites using a single set of credentials (username and password)

Supplier: A person or organization that provides goods or services needed by another organization.

Vendor: A person or company offering goods or services for sale.

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ADS Pitch Document Sample

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